

CHAPTER

33



Skin Diseases

Learning Objectives

- ⇒ To have brief introduction about skin disease
- ⇒ To describe clinical features of allergic disorders – eczema, urticaria
- ⇒ To describe the clinical features of squamous lesions – psoriasis, lichen planus
- ⇒ To know bullous lesions – pemphigus, pemphigoid, mycotic skin diseases
- ⇒ To know about leprosy, vitiligo and cellulitis
- ⇒ To do the necessary physical examinations to be performed in above mentioned diseases
- ⇒ To perform the necessary laboratory investigations in above mentioned diseases and their interpretations

Introduction

History and physical examination are adequate for diagnosing many of the skin lesions. Some require biopsy or other testing. Important information from the history includes

- ⊗ Personal or family history of atopy (suggesting atopic dermatitis)
- ⊗ Occupational exposures (contact dermatitis)
- ⊗ Long term exposure to sunlight or other forms of radiation (benign and malignant skin tumours)
- ⊗ Systemic diseases (diabetes and candida infection or tinea infection)
- ⊗ Sexual history (syphilis and gonorrhoea)
- ⊗ Use of drugs (Steven Johnson syndrome, toxic epidermal necrolysis)
- ⊗ As far as physical examination of the patient is concerned, the main focus is on determining primary and secondary lesions.
- ⊗ In the examination of the skin, the patient should be assessed prior to the detailed clinical history. The observed clinical findings should be integrated to the relevant historical information.
- ⊗ Skin examination is one of the easiest examination to be performed and can be performed without the need for complex investigations, although a biopsy may be needed to confirm the diagnosis.

⊗ An ideal skin examination includes evaluation of skin, hair, nails and mucosal surfaces of mouth, eyes, nose, nasopharynx and anogenital region. During the initial examination of the patient, it is important that the patient is disrobed as completely as possible. This will minimise the chance of missing the important individual skin lesion. The patient should be first viewed from a distance of 1.5-2 m (4-6 feet) so that general character of the skin and distribution of the lesions can be evaluated.

⊗ If the person is presented with skin lesion localised to one particular part of the body, usually the cause of the skin lesion is usually exogenous and if the person is presented with generalised skin lesions usually the cause of the skin lesion is endogenous. If the person is having linear skin lesion, usually the cause of the lesion is either surgical or use of sharp instrument like knife can be predicted. If the person is having the skin lesion with irregular surface, probable cause of the lesion could be exogenous such as road traffic accident or sports injury.

⊗ **Four basic features of skin lesion** must be noted and considered during the physical examination.

1. **Distribution of the eruption** – Which part of the body is afflicted? Among the integumentary system, whether nail or hairs or skin or mucosa afflicted? These informations should be analysed.
2. **Types of primary and secondary lesions** – One should

have the thorough knowledge of primary and secondary lesions and their distribution in the body should be properly analysed. Each dermatological illness is usually having typical primary and secondary lesions in specific location of the body.

3. **Shape of individual lesions** – It should be analysed so that diagnosis of the dermatological illness can be accurately made. For example, the presence of wheal in the body suggests the allergic dermatitis.
4. **Arrangement of the lesions** – Arrangement of the lesion in the different parts of the body will also aid in the diagnosis of the disease.

Eczema

The term eczema is often used to denote dermatitis (inflammation of dermis). This is superficial inflammation of the skin characterized by redness, oedema, oozing, crusting, scaling and sometimes formation of vesicles. This can occur in several forms: atopic dermatitis, contact dermatitis, exfoliative dermatitis, hand and foot dermatitis, stasis dermatitis, seborrheic dermatitis. In the following pages, we are going to learn in brief about above said diseases.

Atopic dermatitis

This is an immune mediated inflammation of the skin arising from an interaction between genetic and environmental factors. Pruritus (itching) is the primary symptom, skin lesions range from mild erythema (redness) to severe lichenification. Diagnosis is done by history and physical examination.

- ⊗ This disease is linked to proallergic/ proinflammatory T cell immune responses. These responses are attributable to cleaner indoor environments, trend toward smaller families, early use of vaccinations and antibiotics deprive children of early exposure to infections and allergens that otherwise suppress proallergic T cells and thereby induce tolerance to various antigens.
- ⊗ It can be either IgE mediated (70-80%) of the cases; intrinsic (20-30%) of the cases. Extrinsic variety occurs due to different triggers such as foods (milk, eggs, soy, wheat, peanuts, fish); airborne allergens (dust mites, molds); staphylococcus aureus colonization on skin due to deficiencies in endogenous antimicrobial peptides); topical products (cosmetics). Intrinsic variety is having idiopathic cause as far as manifestation of atopic dermatitis is concerned.
- ⊗ In the acute phase, which last for 1 to 2 months, red, weeping, crusted lesions appear on the face and spread to the neck, scalp, extremities and abdomen. In the chronic phase, scratching and rubbing create skin lesions (typically erythematous macules and papules that lichenify with continuous scratching).

Lesions typically appear in antecubital and popliteal fossae and on the eyelids, neck and wrists and may occasionally become generalized. Lesions slowly resolve to dry scaly macules (xerosis) that can fissure and facilitate exposure to irritants and allergens. In older children and adults, intense itching is the common feature.

Contact dermatitis

This is acute inflammation of the skin caused by irritants or allergens. Many substances can cause this reaction such as cosmetics, fragrances, jewelry, plants and detergents etc. The primary symptom is pruritus. Skin changes range from erythema to blistering and ulceration, often on or near the hands but occurring on any exposed skin surface. Diagnosis is by exposure history, examination and sometimes skin patch testing.



Figure 128: Atopic dermatitis

Exfoliative dermatitis



Figure 129: Exfoliative dermatitis

This is widespread erythema and scaling of the skin

caused by preexisting skin disorders, drugs, cancer or unknown causes. Symptoms and signs are pruritus, diffuse erythema and epidermal sloughing. Diagnosis is clinical.

Hand and foot dermatitis

Hand and foot dermatitis is not a single disorder. Rather, it is a categorization of dermatitis that affects the hands and feet selectively due to one of several causes. Patients often present with isolated dermatitis of the hands or feet. Causes include contact dermatitis, fungal infection, psoriasis, scabies. Other causes include systemic viral infection in children (hand-foot-mouth disease). Some causes are idiopathic. Diagnosis is inferred from location and appearance of the skin lesions.



Figure 130: Hand and foot dermatitis

Stasis dermatitis

This is inflammation of the skin of the lower limbs caused by chronic venous insufficiency. Symptoms are itching, scaling, hyperpigmentation, and sometimes ulceration. Diagnosis is clinical.



Figure 131: Stasis dermatitis

Seborrheic dermatitis

This is inflammation of skin that has a high density of sebaceous glands (face, scalp, upper trunk). The cause is unknown, but *Pityrosporum ovale*, a normal skin organism play some role. This is occurring in patients

with HIV and in those with certain neurological disorders. The patient will have occasional pruritus, dandruff, and yellow, greasy scaling along the hairline and on the face. Diagnosis is made by examination.



Figure 132: Seborrheic dermatitis

Urticaria

Urticaria is migratory, well-circumscribed, erythematous, pruritic plaques on the skin. This may also be accompanied by angioedema which results from mast cell and basophil activation in the deeper dermis and subcutaneous tissues and manifests as oedema of the face and lips, extremities or genitals. Angioedema can be life threatening if airway obstruction occurs because of laryngeal oedema or tongue swelling.

- Urticaria results from release of histamine, bradykinin, kallikrein, and other vasoactive substances from mast cells and basophils in the superficial dermis, resulting in intradermal oedema caused by capillary and venous vasodilatation and occasionally caused by leukocyte infiltration.
- Caused by hypersensitivity reactions, autoimmune causes.
- The main clinical symptom is itching associated with redness which leads to the occurrence of wheals which will get subsided after due course of time.
- History of present illness should include a detailed account of the individual episodes of urticaria, including distribution, size and appearance of lesions; frequency of occurrence; duration of individual lesions; and any prior episodes. Activities and exposures during, immediately before and within the past 24 hours of the appearance of urticaria should be noted. Clinicians specifically should ask about recent exercise, exposure to potential allergens, insects, or animals or new foods; new laundry detergent or soap; recent infections. Important associated symptoms include rhinorrhoea, swelling of the face and tongue and dyspnoea.
- Review of systems: should seek symptoms of causative disorders, including fever, fatigue, abdominal pain, and diarrhoea (infection); heat or cold intolerance.

tremor or weight change (autoimmune thyroiditis); joint pain (SLE); malar flush (SLE); dry eyes and dry mouth (Sjogren's syndrome); cutaneous ulcers and hyperpigmented patches after resolution of rashes (urticarial vasculitis).

- ❖ Past medical history includes a detailed allergy history (allergies, asthma, eczema); and possible known possible causes (autoimmune disorders and cancer). Family history should elicit any history of rheumatoid disease, autoimmune disorders or cancer.
- ❖ Social history should cover any recent travel, and any risk factors for transmission of infectious disease (e.g. hepatitis and HIV).
- ❖ **Physical examination:** Vital signs should note the presence of bradycardia or tachycardia and tachypnoea. General examination should immediately seek any signs of respiratory distress.
- ❖ Examination of the head should note any swelling of the face, lips or tongue; scleral icterus; malar rash or dry eyes and dry mouth. The oropharynx should be inspected and the sinuses should be palpated for detecting occult infection.
- ❖ Abdominal examination should note any masses, hepatomegaly, splenomegaly or tenderness. Neurological examination should note the presence of tremor or hyperreflexia or hyporeflexia. Musculo skeletal examination should note the presence of inflamed or deformed joints. Skin examination should note the presence and distribution of urticarial lesions as well as cutaneous ulceration.
- ❖ Dermatographism (see Figure 133) is the sign which confirms the diagnosis of urticaria.



Figure 133: Urticaria

- ❖ Angioedema (see Figure 134) (swelling of the face, lips or tongue); stridor (inspiratory crowing type of sound) or other respiratory distress; hyperpigmented lesions, ulcers or urticaria that are present for > 48 hours; signs of systemic illness such as fever, jaundice, cachexia or lymphadenopathy may imply the severity of the disease.
- ❖ The patient has to be managed at the earliest.



Figure 134: Angioedema

Psoriasis

Psoriasis is an inflammatory disease that manifests most commonly as well-circumscribed, erythematous papules and plaques covered with silvery scales. Cause is unclear but seems to involve immune system.

Common triggers include trauma, infection and certain drugs. Symptoms are usually minimal with occasional mild itching, but cosmetic implications may be major. Some people develop severe disease with arthritis. Diagnosis is by appearance and distribution of the lesions.



Figure 135: Psoriasis

- ❖ Psoriasis is hyperproliferation of epidermal keratinocytes combined with inflammation of the epidermis and dermis. It affects about 1 to 5% of the population worldwide; peak onset is seen at ages between 16 to 22 and 57 to 60 years, but the disorder can occur at any age.

- Cause is unclear. Some of the triggers such as injury (Koebner's phenomenon), sunburn, HIV, emotional stress, alcohol consumption, β haemolytic streptococcal infection, drugs are considered to worsen the clinical condition.
- Lesions are asymptomatic or pruritic and are often localized on the scalp, extensor surfaces of the elbows, knees, sacrum, buttocks, penis. The nails, eyebrows, axillae, umbilicus and perianal region may also be affected. The disease can be widespread, involving confluent areas of the skin extending between these regions. Lesions differ in appearance depending on type.
- Lesions are discrete, erythematous papules or plaques covered with thick, silvery, shiny scales. Lesions appear gradually and remit and recur either spontaneously or with appearance and resolution of the triggers.
- Arthritis develops in 5 to 30% of the cases and can be disabling; joint destruction may ultimately occur.
- Psoriasis is rarely life threatening disease, but can affect a patient's self image.
- Diagnosis is by clinical evaluation and rarely biopsy is performed to make the diagnosis. Differential diagnosis includes seborrheic dermatitis, dermatophytosis, cutaneous lupus erythematosus, eczema, lichen planus, and secondary syphilis.

Lichen Planus

Lichen planus (LP) is a recurrent, pruritic, inflammatory eruption characterized by small, discrete, polygonal, flat-topped, violaceous papules that may coalesce into rough scaly patches, often accompanied by oral lesions. Diagnosis is usually clinical and supported by skin biopsy.



Figure 136: Lichen planus

- LP is thought to be caused by T cell mediated response against basal epithelial keratinocytes in people with genetic predisposition.
- Typically lesions are pruritic, purple, polygonal, flat-topped papules and plaques. Lesions initially are 2 to 4 mm in diameter, with angular borders, a violaceous colour and a distinct sheen in cross lighting. They are usually symmetrically distributed, moist commonly on flexural surfaces of the wrists, legs, trunk, glans penis and oral and vaginal mucosae, but can be widespread. The face is rarely involved. Onset may be abrupt or gradual.
- During acute phase, new papules may appear as sites of minor skin injury (Koebner phenomenon), such as superficial scratch. Lesions may coalesce or change over time, becoming hyperpigmented, atrophic, hyperkeratotic or vesiculobullous. Although pruritic, lesions are rarely excoriated or crusted. If the scalp is affected, patchy scarring alopecia may occur.
- The oral mucosa is involved in 50% of the patients; oral lesions may occur without cutaneous lesions and usually persist for life. Reticulated, lacy, bluish white, linear lesions (Wicham's striae) are a hallmark of oral LP, especially on oral mucosae. Tongue margins and gingival mucosae in edentulous areas may also be affected. Chronic exacerbations and remissions are common.
- Vulvar and vaginal mucosae are often involved in upto 50% of the women. In men, genital involvement is common, especially in glans penis.
- Nails are involved in 10% of the patients. Findings vary in intensity with nail bed discolouration, longitudinal ridging and lateral thinning, and complete loss of nail matrix and nail, with scarring of the proximal nail fold onto nail bed.
- Diagnosis is by clinical evaluation and skin biopsy.
- **Prognosis:** Many cases resolve without intervention, presumably because the inciting agent is no longer present. Recurrence after years may be due to reexposure to the trigger or some change in the triggering mechanism. Sometimes, treatment of a previously occult infection, such as a dental abscess, result in resolution. Vulvovaginal LP may be chronic and refractory to therapy, causing decreased quality of life.

Pemphigus

Pemphigus vulgaris is an uncommon, potentially fatal, autoimmune disease characterized by intraepidermal blisters and extensive erosions on an apparently healthy skin and mucous membranes. Pemphigus foliaceus is a generally benign blistering disorder characterized by splitting high in epidermis, causing erosions to form on the skin. These are categorized under bullous diseases. In

this section, we are going to discuss pemphigus vulgaris as this is most fatal, commonly observed in Ayurveda hospitals among the blistering diseases.

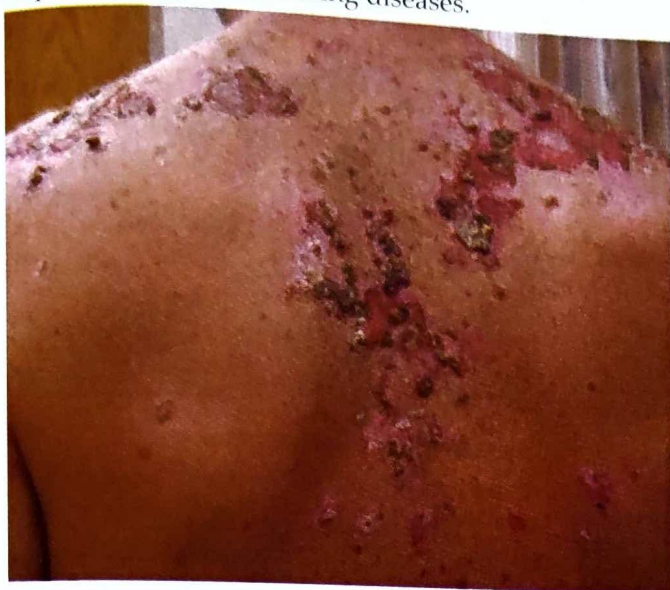


Figure 137: Pemphigus

- ❖ Pemphigus vulgaris (PV) occurs in middle aged or elderly parents and is rare in children. It is characterized by the presence of autoantibodies directed against intercellular adhesions molecules. These autoantibodies are present both in skin and serum.
- ❖ The primary lesions are flaccid bullae of various sizes, but often skin or mucosa just shears off, leaving painful erosions. Lesions typically occur first in the mouth, where they rupture and remain as chronic, often painful, erosions for variable periods before the skin is affected; dysphagia and poor oral intake are common.
- ❖ Lesions also appear on the upper oesophagus.
- ❖ Cutaneous bullae typically arise from normal appearing skin, rupture and leave raw area and crusting. Itching is usually absent. Open skin lesions often become infected. If large portions of the body are affected, fluid and electrolyte loss may be significant. Nikolsky's sign is positive in the patients of pemphigus vulgaris (PV).
- ❖ Diagnosis is by clinical evaluation; biopsy with direct immunofluorescence testing.
- ❖ As far as prognosis is concerned, before systemic corticosteroids were used, pemphigus vulgaris was usually fatal, most patients died within 5 years of onset. Even with treatment, PV is a serious disorder with an inconsistent and unpredictable response to the treatment, a prolonged course, and virtually inevitable adverse drug effects.

Pemphigoid

Bullae, are elevated, fluid filled blisters ≥ 5 mm in

diameter. Bullous pemphigoid is an autoimmune, skin disorder causing chronic, pruritic, bullous eruptions in elderly patients. Diagnosis is done by skin biopsy.



Figure 138: Bullous pemphigoid

- ❖ In bullous pemphigoid (BP), antibodies are directed against the base membrane zone of the epidermis, causing separation between the epidermis and dermis. Bullous pemphigoid must be differentiated from pemphigus vulgaris.
- ❖ Characteristic tense bullae develop on normal appearing or erythematous skin, most often in flexural areas. Nikolsky's sign, in which lateral pressure on skin adjacent to a blister causes epidermal detachment is negative.
- ❖ BP can manifest initially as hives with annular, dusky red, oedematous lesions with or without peripheral vesicles. Itching is common, usually without other symptoms. Oral lesions occur in about one third of patients but heal rapidly.
- ❖ Diagnosis is by biopsy and antibody titers.
- ❖ Prognosis is good, and the disorder usually subsides within months to years; however the disease is fatal, especially in the elderly and debilitated patients, with death being caused by infection and sepsis or the effects of the drugs.

Mycotic Skin Diseases

The term mycosis or fungal infections refers to the diseases caused by fungi (yeast or mold). Dermatophytosis is the term used for fungal infections of the keratin in the skin and nails. Symptoms and signs may vary by the site of the infection. Diagnosis is done by clinical appearance and by examination of skin scraping on potassium hydroxide wet mount.

- ❖ Dermatophytes are molds that require keratin for nutrition and must live on stratum corneum, hairs on nails to survive. Human infections are caused by *Epidermophyton*, *Microsporum* and *Trichophyton* spp. These infections are different from candidiasis in that they are rarely if ever invasive. Transimission

is person to person, animal to person, and rarely soil to person.

- ⊗ Symptoms and signs vary by site (skin, hair, nails). Organism virulence and host susceptibility and hypersensitivity determine severity. Most often, there is little or no inflammation; asymptomatic or mildly itching lesions with a scaling, slightly raised border remit and recur intermittently.
- ⊗ Occasionally, inflammation is more severe and manifests as sudden vesicular or bullous disease (usually of the foot) or as an inflamed boggy lesion of the scalp (kerion).
- ⊗ Diagnosis is done by clinical appearance and potassium hydroxide wet mount. Differential diagnosis includes folliculitis decalvans, bacterial pyoderma and entities causing scarring alopecia.
- ⊗ Several terms are used for mycotic skin infections depending upon the location.
- ⊗ Tinea barbae (infection of the beard); tinea capitis (infection of the scalp); tinea corporis (infection of the face, trunk and extremities); tinea cruris (infection of the groin); tinea pedis (infection of the feet). All these infections are caused by dermatophytes.



Figure 139: Mycotic skin disease

- ⊗ Tinea versicolor (pityriasis versicolor) is skin infection caused by *Malassezia furfur* that manifests as multiple asymptomatic scaly patches varying in colour from white to brown. Diagnosis is based on clinical appearance and skin scrapings.
- ⊗ Classically, it causes the appearance of multiple tan, brown, salmon, or white scaling lesions on the trunk, neck, abdomen and occasionally face. The lesions coalesce in some of the occasions.
- ⊗ Diagnosis is clinical appearance and by potassium hydroxide wet mount.



Figure 140: Tinea versicolor

Leprosy

Leprosy is a chronic infection caused by the acid fast bacillus *Mycobacterium leprae*, which has a unique tropism for peripheral nerves, skin and mucous membranes. Symptoms are myriad and include anaesthetic, polymorphic skin lesions and peripheral neuropathy. Diagnosis is done by clinical evaluation and confirmed by biopsy.

- ⊗ Leprosy can develop at any age but appears most often in people aged 5 to 15 years or after 30 years.
- ⊗ Humans are the main natural reservoir for *M. leprae*. The mechanism of spreading of leprosy is unclear. It is thought to be passed from person to person through nasal droplets and secretions. Casual contact (e.g. simply touching someone with the disease) and short term contact may not seem to spread the disease.
- ⊗ *M. leprae* grow slowly (doubling in 2 weeks), the usual incubation period ranges from 6 months to 10 years. Once infection develops, haematogenous dissemination can occur.

Classification

- ⊗ Leprosy can be categorized by type and number of lesions observed in the skin.
 - **Paucibacillary:** ≤ 5 skin lesions with no bacteria detected on samples from the affected areas
 - **Multibacillary:** ≥ 6 skin lesions, bacteria detected on samples from the skin lesions or both
- ⊗ Leprosy can also be classified by cellular response and clinical findings.
 - Tuberculoid
 - Lepromatous
 - Borderline
- ⊗ People with tuberculoid leprosy typically have strong cell mediated response, which limits the

disease to few skin lesions (paucibacillary). People with lepromatous or borderline leprosy typically have poor cell mediated immunity and have more severe, systemic infection with widespread bacterial infiltration of skin, nerves and other organs (nose, testes, kidneys). They have more skin lesions (multibacillary) and the disease is more contagious.

- ⊛ Symptoms usually do not begin until > 1 year after infection (average 5 to 7 years). Once the symptoms begin, they progress slowly. Leprosy mainly affects the skin and peripheral nerves. Nerve involvement causes numbness and weakness in areas controlled by the affected nerves.

- **Tuberculoid leprosy:** Skin lesions consist of one or few hypoesthetic, centrally hypopigmented macules with sharp, raised borders. The rash, as in all forms of leprosy, is non pruritic. Areas affected by this rash are numb because of damage to the underlying nerves, which may be palpably enlarge.



Figure 141: Tuberculoid leprosy

- **Lepromatous leprosy:** Much of the skin and many areas of the body, including the kidneys, nose and testes, may be affected. Patients have macules, papules, nodules and plaques, often symmetric. Peripheral neuropathy is more severe than in tuberculoid leprosy, with more areas of numbness.
- **Borderline leprosy:** Features of both tuberculoid and lepromatous leprosy are present. Without treatment, borderline leprosy may be less severe and more like tuberculoid form, or it may worsen and become more like the lepromatous form.

Complications

- ⊛ The most severe complications result from the peripheral neuropathy, which causes deterioration of the sense of touch and a corresponding inability to feel pain and temperature. Patients may unknowingly feel pain and temperature. Repeated burn, cut or otherwise harm themselves. Repeated damage may lead to loss of digits. Muscle weakness

can result in deformities (e.g. clawing of 4th and 5th fingers caused by ulnar involvement, foot drop caused by peroneal nerve involvement). Papules and nodules can be particularly lead to disfiguration of the face.

- ⊛ Some other complications observed in the patients of leprosy are plantar ulcers, erosion and collapse of nasal septum, erectile dysfunction, amyloidosis and renal failure.
- ⊛ Diagnosis is done by microscopic examination of skin biopsy specimen.

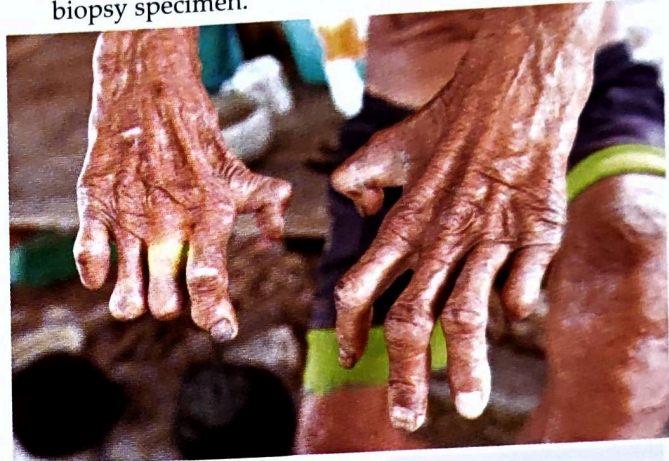


Figure 142: Clawing of hands in ulnar involvement of leprosy

Vitiligo

Vitiligo is a loss of skin melanocytes that causes the areas of skin depigmentation of varying sizes. Cause is unknown. But the condition may be autoimmune, up to 1/3rd of the patients have evidence of other autoimmune disease. Diagnosis often obvious on examination.

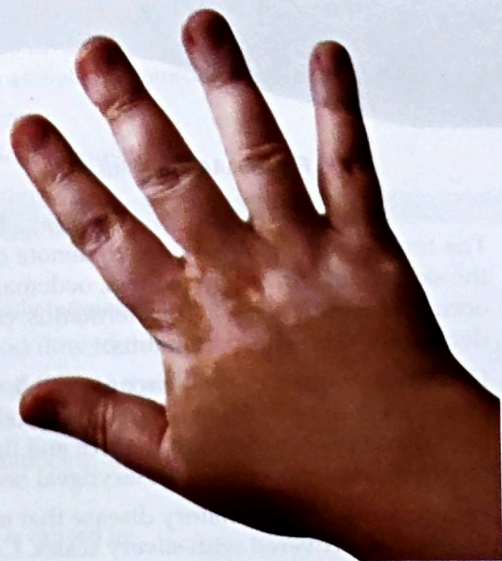


Figure 143: Vitiligo

- ⊛ Aetiology is unclear, but melanocytes are lacking in

affected areas.

- This disease is characterized by depigmented areas, usually sharply demarcated and often symmetric. Depigmentation may be localized involving one or two spots or entire body segments (segmental vitiligo), rarely it may be generalized, involving most of the skin surface.
- Vitiligo most often involves face (especially around the orifices), digits, dorsal hands, flexor wrists, elbows, knees, shins, dorsal ankles, armpits, inguinal areas, anogenital area, umbilicus and nipples. Hair in vitiliginous area is usually white.
- Diagnosis is done by clinical evaluation. Differential diagnosis includes postinflammatory hypopigmentation, leprosy, chemical leukoderma and leukoderma due to melanoma.

Cellulitis

Cellulitis is acute bacterial infection of the skin and subcutaneous tissue most often caused by streptococci or staphylococci. Symptoms and signs include pain, rapidly spreading erythema, oedema, fever and enlargement of regional lymph nodes. Diagnosis is done by appearance of the skin lesions, culture of the tissue of the lesion.



Figure 144: Cellulitis

- Cellulitis is most often caused by *Streptococcus pyogenes* or *Staphylococcus aureus*. Risk factors for the development of the lesion include trauma, ulceration, fungal infection, other skin barrier compromise due to preexisting skin disease. Persons with chronic venous insufficiency or lymphoedema may also contribute the rapid spread of cellulitis of the susceptible part of the body.
- Infection is most commonly observed in the lower limb. Cellulitis is typically unilateral in presentation. Stasis dermatitis mimics cellulitis, but is usually bilateral in clinical presentation. Local erythema and local tenderness, lymphangitis and lymphadenitis are observed. The skin of the affected area is usually hot, red and oedematous, often with surface of the skin appears as the skin of the orange (peau d orange). The borders of the cellulitis are indistinct. Petechiae are common, large areas of ecchymoses are rare. Vesicles and bullae may develop and rupture, occasionally with necrosis of the involved skin.
- Fever, chills, tachycardia, headache, hypotension and delirium may precede cutaneous findings for several hours.
- Diagnosis is done by physical examination, blood and tissue culture.
- Differential diagnosis of cellulitis include stasis dermatitis and deep venous thrombosis. Stasis thrombosis is usually bilateral, whereas cellulitis is unilateral in clinical presentation. Deep vein thrombosis is having several distinctive clinical features than cellulitis i.e. normal skin temperature, normal or cyanotic skin colour, smooth surface and non exsistant lymphangitis and lymphadenitis. Whereas cellulitis is presented with raised skin temperature, reddish discolouration of the skin, peau de orange skin appearance and frequent lymphangitis and lymphadenitis.
- **Prognosis:** most of the cellulitis will respond with antibiotic therapy.

Key summary of the chapter

- The term eczema is often used to denote dermatitis (inflammation of dermis). This is superficial inflammation of the skin characterized by redness, oedema, oozing, crusting, scaling and sometimes formation of vesicles. This can occur in several forms: atopic dermatitis, contact dermatitis, exfoliative dermatitis, hand and foot dermatitis, stasis dermatitis, seborrheic dermatitis.
- Urticaria is migratory, well-circumscribed, erythematous, pruritic plaques on the skin. This may also be accompanied by angioedema which results from mast cell and basophil activation in the deeper dermis and subcutaneous tissues obstruction occurs because of laryngeal oedema or tongue swelling.
- Psoriasis is an inflammatory disease that manifests most commonly as well-circumscribed, erythematous papules and plaques covered with silvery scales. Cause is unclear but seems to involve immune system. Common triggers include trauma, infection and certain drugs. Symptoms are usually minimal with occasional mild itching, but cosmetic implications may be major. Some people develop severe disease with arthritis. Diagnosis is by appearance and distribution of the lesions.

- Lichen planus (LP) is a recurrent, pruritic, inflammatory eruption characterized by small, discrete, polygonal, flat-topped, violaceous papules that may coalesce into rough scaly patches, often accompanied by oral lesions. Diagnosis is usually clinical and supported by skin biopsy.
- Pemphigus vulgaris is an uncommon, potentially fatal, autoimmune disease characterized by intraepidermal blisters and extensive erosions on an apparently healthy skin and mucous membranes. Pemphigus folicaceous is a generally benign blistering disorder characterized by splitting high in epidermis, causing erosions to form on the skin. These are categorized under bullous diseases. In this section, we are going to discuss pemphigus vulgaris as this is most fatal, commonly observed in Ayurveda hospitals among the blistering diseases.
- In bullous pemphigoid, bullae, are elevated, fluid filled blisters ≥ 5 mm in diameter. Bullous pemphigoid is an autoimmune, skin disorder causing chronic, pruritic, bullous eruptions in elderly patients. Diagnosis is done by skin biopsy.
- The term mycosis or fungal infections refers to the diseases caused by fungi (yeast or mold). Dermatophytosis is the term used for fungal infections of the keratin in the skin and nails. Symptoms and signs may vary by the site of the infection. Diagnosis is done by clinical appearance and by examination of skin scraping on potassium hydroxide wet mount.
- Leprosy is a chronic infection caused by the acid fast bacillus *Mycobacterium leprae*, which has a unique tropism for peripheral nerves, skin and mucous membranes. Symptoms are myriad and include anaesthetic, polymorphic skin lesions and peripheral neuropathy. Diagnosis is done by clinical evaluation and confirmed by biopsy.
- Vitiligo is a loss of skin melanocytes that causes the areas of skin depigmentation of varying sizes. Cause is unknown. But the condition may be autoimmune, up to 1/3rd of the patients have evidence of other autoimmune disease. Diagnosis often obvious on examination.
- Cellulitis is acute bacterial infection of the skin and subcutaneous tissue most often caused by streptococci or staphylococci. Symptoms and signs include pain, rapidly spreading erythema, oedema, fever and enlargement of regional lymph nodes. Diagnosis is done by appearance of the skin lesions, culture of the tissue of the lesion.

Self-Assessment: Questions for Reviews

Short essay questions

- Describe eczema
- Describe atopic dermatitis
- Explain urticaria
- Describe psoriasis
- Describe pemphigus vulgaris
- Describe tinea versicolor
- Explain leprosy
- Describe vitiligo
- Explain cellulitis

MCQs

1. Macule is considered as

- a. Inspection finding;
- b. Palpable finding;
- c. Both inspection and palpation finding;
- d. None of the above

Answer: (a)

2. The layer of the epidermis that contains melanocytes is

- a. Stratum basale;
- b. Stratum lucidum;
- c. Stratum spinosum;
- d. Stratum corneum

Answer: (a)

3. Most body odour is the result of bacterial metabolism of the secretions produced byglands.

- a. Ceruminous glands;
- b. Sebaceous glands;
- c. Eccrine sweat glands;
- d. Apocrine sweat glands

Answer: (d)

4. The layer of the skin that provides protection against bacteria as well as chemical and mechanical injuries is

- a. Dermis;
- b. Subcutaneous tissue;
- c. Epidermis;
- d. Sebum layer

Answer: (c)

5. When body temperature raises above normal

- a. Blood flow to the skin decreases;
- b. Blood flow to the skin increases;
- c. Vasoconstriction occurs;
- d. Melanocyte activity increases

Answer: (b)

6. Leprosy is caused by

- a. Virus;
- b. Bacteria;
- c. Fungus;
- d. Protozoa

Answer: (b)

